

AARUSH GHOSH

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EDUCATION

University of Waterloo

Expected Apr 2028

Bachelor of Statistics & Computational Math with Computer Science Minor, Co-op

- **Relevant Courses:** Object-Oriented Software Development, Data Structures, Algorithms, Data-Intensive Distributed Computing, Networks and Distributed Computer Systems, Statistics, Optimization, Graph Theory, Linear Modeling, Stochastic Processes

EXPERIENCE

Versa Networks

May 2026 - Present

Software Engineer Intern

Santa Clara, United States

- Architected Versa's first end-to-end **UEBA MLOps pipeline**, enabling Kubernetes-native model training and zero-downtime delivery of **200+ models** across **50+ independently managed global clusters**; formalized the design in an **architecture document**
- Productionized model training with **Argo Workflows** and **MLflow**, providing atomic model versioning, observability, and governance to prevent corrupted or regressed models from reaching production
- Implemented the delivery plane as a **Go microservice** integrated as **Helm charts**, using **CronJob-based** reconciliation and **sidecar containers** to safely verify, install, and roll out model releases
- Built a **Redis-backed caching layer** that orchestrates **Kubernetes namespace and pod provisioning** for multi-tenant deployments, improving performance across customer onboarding and offboarding workflows

Qorsa Corp.

Jan 2026 - April 2026

Machine Learning Engineer Intern

Waterloo, Canada

- Architected a distributed **Dgraph** storage backend for a **GraphRAG** engine serving government intelligence contracts in **Python**
- Implemented the full storage interface with **MVCC aware** write paths, batch reads, and graph traversal for up to **100,000+** entities
- Owned the production deployment of the **GraphRAG platform**, delivering GPU-optimized **vLLM inference**, automated **CI/CD**, and **IAM-based service authentication** that eliminated manual credential management
- Built document ingestion endpoints supporting **7+** formats for a knowledge graph pipeline; implemented **VLM-based** extraction with PDF-to-markdown conversion, reducing LLM hallucination **by 68%**

University of Waterloo

Jan 2025 - Apr 2025

Software Developer

Waterloo, Canada

- Automated budget planning for **9,000+ students** using a **Markov chain forecasting model** for the Dean of Mathematics
- Migrated legacy SQL queries to a **PySpark** ETL pipeline on **Databricks**, improving runtime by **78%**

CGI

May 2024 - Aug 2024

AI Developer Intern

Toronto, Canada

- Deployed an enterprise chatbot into production for marketing analytics by building a **RAG pipeline** processing **Google Analytics 4** data, enabling natural language queries over web traffic patterns for an insurance client
- Built an **ETL pipeline** on **Databricks** using the **medallion architecture** and **PySpark** to transform raw GA4 events into weekly aggregates
- Improved **text-to-SQL** accuracy by **23%** through schema **metadata enrichment** and retrieval-augmented parsing with **LlamaIndex**, achieving **89% F1-score** on multi-table marketing analytics queries

PROJECTS & EXTRACURRICULARS

reLive: Concert Video Logging Platform | **McHacks 13 Winner**

- Led backend development in **Go and Gin** to evolve the product from a hackathon win into a deployed app handling **1,000+ concurrent requests** with active users across the university music community, sustaining request latency **under 200ms**
- Designed a worker-pool concurrency model backed by **PostgreSQL** job queue and **goroutines and channels** to parallelize video uploads
- Deployed with **Docker** with automated **CI/CD** through **Artifact Registry**, enabling reliable image versioning and zero-downtime
- Integrated **AWS S3** object storage, using **presigned URLs** for client-to-storage uploads to keep large file transfers off application servers

WAT.ai: Research Developer

- Researched unsupervised anomaly detection within **Canada's largest undergraduate AI hub**, implementing and comparing unsupervised clustering algorithms across **105 IoT devices** to identify network intrusion patterns; presented at **CUCAI 2024**

SKILLS

Languages: Python, Go, SQL, TypeScript, C++, Bash

ML/AI: PyTorch, TensorFlow, vLLM, Ollama, CUDA, MLflow, HuggingFace, LangChain, LlamaIndex

Backend & Data: FastAPI, Gin, gRPC, Django, Express.js, PostgreSQL, MongoDB, Redis, Kafka, Flink, Spark, Neo4j, Airflow, Databricks

Infrastructure & DevOps: AWS, GCP, Azure, Docker, Kubernetes, Helm, GitHub Actions, Git, Linux